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DC Motor Speed Stability Diagnostic Worksheet

For maintenance engineers and machine builders troubleshooting unstable, hunting or load-dependent motor speed

Work through the steps in order and record readings as you go. Steps 1 and 2 usually identify the cause before any controller setting is touched. Isolate and lock out the drive before any hands-on mechanical or commutator inspection.

Step 1 — Capture and classify the fluctuation

- [ ] Record nameplate data: voltage, rated current, rated speed, duty type, and the controller make and model.
- [ ] Log speed, armature current and terminal voltage together over at least five machine cycles, at 1 ms resolution or finer if supply ripple is suspected.
- [ ] Calculate the period of one shaft revolution (60 / rpm seconds) and of one machine cycle; note the 10 ms ripple period of a single-phase rectifier on 50 Hz mains.
- [ ] Classify the pattern: load droop, slow drift, hunting, periodic dip, or random dip.
- [ ] Measure speed at the motor shaft and at the driven shaft with the same instrument and compare.

Step 2 — Supply and voltage headroom

- [ ] Measure voltage at the motor terminals, not at the supply, at no load and at full load.
- [ ] Investigate any drop of more than about 2–3 % between supply and motor terminals at full load: cable size, lugs, fuses, contactors.
- [ ] Record the lowest supply voltage seen in service (battery at cut-off, mains at its lowest point of the shift).
- [ ] Confirm that the armature voltage needed at maximum set speed and full load is below the lowest available controller output voltage.
- [ ] On rectified supplies, check ripple and form factor with an oscilloscope or true-RMS meter.

Step 3 — Open-loop behaviour and drive settings

- [ ] Calculate speed regulation: (no-load rpm – full-load rpm) / full-load rpm × 100.
- [ ] Compare with the motor test sheet; a higher figure suggests extra circuit resistance, worn brushes or weakened magnets.
- [ ] Compare cold and hot no-load speed; a rise of a few per cent on a ferrite PMDC motor is normal.
- [ ] Set IR compensation at full load until no-load and full-load speeds match, then reduce it slightly.
- [ ] Check the current limit against the motor's rated and peak current, and check ramp settings against the load cycle.

Step 4 — PI speed loop tuning sequence

- [ ] Tune with the real load coupled and, where possible, at operating temperature.
- [ ] Set integral gain to zero; raise proportional gain in small steps until the response rings, then halve it.
- [ ] Add integral gain gradually until steady-state error is removed; keep overshoot below about 10–20 %.
- [ ] Re-check at minimum and maximum speed and at the lightest and heaviest load inertia.
- [ ] Enable anti-windup if the drive reaches current limit during load steps.
- [ ] At low speed, confirm the encoder delivers enough edges per sample (counts = 4 × PPR × rpm / 60 × sample time).

- [ ] Record the final gains and firmware version in the machine file.

Step 5 — Mechanical and commutation checks (drive isolated)

- [ ] Belt tension, glazing and oil contamination; pulley alignment.
- [ ] Coupling condition and alignment; output backlash felt by hand.
- [ ] Commutator: high or low bars, flat spots, burn marks at regular bar spacing.
- [ ] Brush length against the minimum mark, free movement in holders, spring force.
- [ ] Lugs and crimps for tightness and discolouration; flex the harness while logging current only where site electrical safety rules allow.
- [ ] For cyclic loads, overlay speed on cam or crank position and estimate the energy swing per cycle.

Fluctuation Pattern Quick Reference

Pattern	Likely cause	First check	Typical fix
Speed falls with load	Open-loop droop	Regulation % from no load to full load	IR compensation or speed feedback
Slow wander over minutes	Supply or battery variation	Terminal voltage logged with speed	Regulated supply, voltage headroom
Light-load speed rises when warm	Magnet flux falls with temperature	Cold vs hot no-load speed	Closed loop, better cooling
Oscillation that changes with gains	PI loop over-tuned	Reduce integral gain and observe	Retune, anti-windup
Jitter at low speed only	Feedback quantisation, cogging	Edges per sample period	Period measurement, higher PPR, more gearing
Dips repeating each revolution	Commutator bar, open coil, runout	Scope armature current	Skim commutator, repair, realign
Locked to machine cycle	Cyclic load torque	Speed vs cam or crank angle	Flywheel, inertia, feed-forward
Motor steady, load varies	Belt slip or compliance	Speed at both ends	Re-tension belt, stiffer coupling

Regulation figures are typical for industrial DC drives and are for guidance only. Confirm limits against the motor test sheet and the controller manual.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.